

In favour of 5G

We speak to one of the leading economists to gather how 5G deployment could happen in the country

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d You mentioned that If you don't regulate an industry, then it helps it grow and innovate quicker. Where do you draw the line, or rather hit the balance of regulation in such a manner that not only ensures security, but also allows for innovation?

Harold: Let me be clear, the role of the government is to, I call it "define property and contract rights". Regulations are not meant to be completely absent and then suddenly put in to be enforced. And there are other types of regulations that governments put in place as well. In the US we have the Universal Service Fund. We do things to help build up the communications network in rural and remote areas. What I think is that it's important to have rules that are predictable and enforceable over time. Those can include security, so that we can be sure that network equipment is not easily hacked into. That's, I think, entirely consistent with the rule of law. What's remarkable about the development of the wireless industry, and the internet in general, has been the large absence of major governmental players. Generally speaking, wireless carriers are not government owned, equipment manufacturers are not government owned. It's an industry that has developed in response to price signals in the market, in response to extraordinary interest in those types of innovations. The benefits of that have redounded not just to wealthy individuals but to the poorest of the poor. I think it's had a very major effect in India and around the world.



Harold W. Furchtgott-Roth
Economist. Ex-Commissioner of FCC

d The spectrum in India is very expensive, so would you believe a different pricing model for the spectrum would be beneficial? More on the lines of OpEx rather than CapEx?

Harold: Auctions are a good way to assign licences or resources when there are mutually competing applications. As an economist, I strongly hold the view that the purpose of wireless auctions should not be to raise money for the government, it should be to get spectrum out and in use. Because every year that spectrum isn't put to use is something you can never get back. It's a completely renewable resource and if you don't use it per year, that year is lost forever. So a lot of this has to do with how many companies want to come in and provide services, and how the government decides who wants to organise the blocks of spectrum.

d Most of the deployments that we are seeing for 5G are happening in developed economies. They have a lot of opportunities and different industries to be accelerators for the adoption of 5G. In India, We saw 4G being adopted purely because it was cheap. However, given the cost of 5G consumer smartphones we might not see a similar adoption. So in a developing nation, what do you foresee as the key accelerator for adoption?

Harold: Every time there's been a new wireless technology, there have been

people who have said – Do we really need this? It's going to be too costly to deploy new networks, why don't we just stick with what we've got because that's already great. But every time new wireless technologies have been deployed, it turns out, people love it. We have yet to reach a point where someone has deployed a new technology of wireless services, and no one showed up. It just hasn't happened. And I don't think that's going to happen with 5G either. Yes, of course, there may be timing issues. And once one company deploys 5G, I think the other companies are going to deploy 5G because customers are going to want it. I don't think this is a developed country versus a developing country issue. This has happened with every generation of wireless technology. It's not that every country deploys at the same time, there can be wide variations. But it's also not the case that new technologies just completely bypass a country. I just don't think that's going to happen.

Some of the high frequency technologies, what's called millimetre wave, is a very heavily urban type of technology. And for that, there can be an urban-rural divide. You may not see it out in the countryside but there will be 5G technology in the rural areas, it just might not be millimetre wave.

d Doesn't that mean that telecom operators not only have to procure millimetre wave spectrum, but also mid band spectrum or perhaps low-band spectrum. That doubles the expenditure, right?

Harold: I can speak for the United States. In the US, the largest auction proceeds for AWS-3 (Advanced Wireless Services), which was essentially 4G, they raised \$40-50 billion. I don't remember the exact figures. And prices were in the kind of in the \$2 per MHz-POP range, which is pretty astronomical. The millimetre wave band auctions have been generating some revenue but more in the \$2 billion range. Carriers the United States will have some millimetre wave, and have some mid-band spectrum and some low-band spectrum. And they're going to have networks that kind of combine all of those. And I suspect it will be the same here in India. **d**